

Chapter 1 Logic of Compound Statements
Statements and Logical form

Definition 1.1

A **statement** or **proposition** is a declarative sentence that is either true or false, but not both.

Definition 1.2

The **truth value** of a proposition is true (T), if it is a true proposition and false (F), if it is a false proposition.

Example 1.1

p : The year 1973 was a leap year.
is a proposition readily decidable as false.

Note the use of the label ' p :', so that the overall statement is read ' p is the statement: "The year 1973 was a leap year" '. So we use p, q, r, s and t to represent statements and these letters are called **statement variables**, that is, variables that can be replaced by statements.

Example 1.2

Determine whether the following sentences are statements or not. If it is a statement, determine its truth value.

- (a) Selangor is a state in Malaysia.
- (b) The sun rises in the West.
- (c) She is a computer science major.
- (d) $128 = 2^6$.
- (e) $x = 2^6$.
- (f) Is $(2^{10} - 1)$ an even integer?

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(g) Take the book.

(h) $x^2 + x + 1 = 0$, x is a real number.

(i) $x^2 + x + 1 = 0$, x is a complex number.

(j) Maths is fun

Definition 1.3

A table that gives the truth values of a compound statement in terms of its component parts is called a **truth table**.

Connectives

Most mathematical statements are combinations of simpler statement formed through some choice of the words **not**, **and**, **or**, **if...then...**, and **if and only if**. These are called **logical connectives** (or simply **connectives**) and are denoted by the following symbols:

$\sim$ or $\neg$	Not
$\wedge$	And
$\vee$	Or
$\Rightarrow$	If ____, then ____
$\Leftrightarrow$	If and only if

Compound statements

Definition 1.4

- (1) A statement represented by a single statement variable (without any connective) is called a **simple** (or **primitive**) **statement**.
- (2) A statement represented by some combination of statement variables and connectives is called a **compound statement**.

Example 1.3

- (1) A dog or a car is an animal.
- (2) A dog is not an animal.
- (3) $5 < 3$.
- (4) If the earth is flat, then $3 + 4 + 7$.

Truth table

A **truth table** displays the relationships between the truth values of statements. Truth tables are especially valuable in the determination of the truth values of statements constructed from simpler statements

Definition 1.5

If p is a statement variable, the **negation** of p is “not p ” or “it is not the case that p ” and is denoted $\sim p$. It has opposite truth value from p .

Truth Table for $\sim p$

p	$\sim p$
T	F
F	T

Example 1.4

Give the negation of the following statements:

- (a) p : The integer 10 is even.
- (b) q : $2 + 3 > 1$
- (c) r : $3 + 7 = 10$

Definition 1.6

If p and q are statement variables, the **conjunction** of p and q is “ p and q ”, denoted $p \wedge q$. The compound statement $p \wedge q$ is true when both p and q are true; otherwise, it is false.

Truth Table for $p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example 1.5

Form the conjunction of p and q for each of the following:

- (a) p : It is snowing. q : I am cold

- (b) p : $2 < 3$. q : $-5 > -8$

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Example 1.6

Determine the truth value of the following statements:

- (a) $3 < 5$ and $5 + 6 \neq 11$
- (b) 5 is positive and Kuala Lumpur is in Malaysia.
- (c) The integer 2 is even but it is a prime number.

Definition 1.6

If p and q are statement variables, the **disjunction** of p and q is “ p or q ”, denoted $p \vee q$. The compound statement $p \vee q$ is true if at least one of p or q is true; it is false when both p and q are false.

Truth Table for $p \vee q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Note

The notation for inequalities involves “and” and “or” statements:

Let a , b and c be particular real numbers.

$a \leq b$ means $a < b$ **or** $a = b$

$a < b < c$ means $a < b$ **and** $b < c$.

Note

$\sim$ is an unary operation, while $\vee$ and $\wedge$ are binary operations.

Example 1.7

Form the disjunction of p and q for the following:

p : 2 is a positive integer. q : $\sqrt{2}$ is a rational number.

Example 2.8

Determine the truth value of the given statements:

- (a) 3 or -5 is negative.
- (b) $\sqrt{2}$ or π is an integer.
- (c) $5 \leq 5$

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Definition 1.7

A **statement form** is an expression made up of statement variables and connectives.

The **truth table** for a given statement form displays the truth values that correspond to the different combinations of truth values for the variables.

Note

If a statement form s contains n statement variables, there will need to be 2^n rows in the truth table for s .

Example 1.9

$s: p \vee (q \wedge (p \vee r))$ involves 3 statement variables, p , q and r . So there are altogether 2^3 or 8 possible combinations of truth values for p , q and r .

Truth Table for s (s involves 2 statement variables)

p	q	s
T	T	
T	F	
F	T	
F	F	

Truth Table for s (s involves 3 statement variables)

p	q	r	s
T	T	T	
T	T	F	
T	F	T	
T	F	F	
F	T	T	
F	T	F	
F	F	T	
F	F	F	

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Example 1.10

Construct a truth table for the statement form $(p \wedge q) \vee \sim r$.

Steps:

- (1) Set up columns labeled p , q , r , $\sim r$, $(p \wedge q)$ and $(p \wedge q) \vee \sim r$.
- (2) Fill in the p , q and r columns with all the logically possible combinations of T's and F's.
- (3) Use the truth tables for $\sim$ and $\wedge$ to fill in the $\sim r$ and $(p \wedge q)$ columns with the appropriate truth values.
- (4) Finally, fill in the $(p \wedge q) \vee \sim r$ column by considering truth values for $(p \wedge q)$ and $\sim r$.

Truth table for $(p \wedge q) \vee \sim r$

p	q	r	$\sim r$	$(p \wedge q)$	$(p \wedge q) \vee \sim r$
T	T	T	F	T	T
T	T	F	T	T	T
T	F	T	F	F	F
T	F	F	T	F	T
F	T	T	F	F	F
F	T	F	T	F	T
F	F	T	F	F	F
F	F	F	T	F	T

Example 1.11

Construct a truth table for the following statement forms:

- (i) $(\sim p \vee q) \wedge (\sim r)$
- (ii) $(\sim p \wedge \sim r) \vee (r \wedge q)$
- (iii) $q \wedge \sim(\sim p \vee r)$

(i) **Truth table for $(\sim p \vee q) \wedge (\sim r)$**

p	q	r				
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

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(ii) Truth table for $(\sim p \wedge \sim r) \vee (r \wedge q)$

p	q	r					
T	T	T					
T	T	F					
T	F	T					
T	F	F					
F	T	T					
F	T	F					
F	F	T					
F	F	F					

(iii) Truth table for $q \wedge \sim(\sim p \vee r)$

p	q	r					
T	T	T					
T	T	F					
T	F	T					
T	F	F					
F	T	T					
F	T	F					
F	F	T					
F	F	F					

Tautology, Contradiction and Contingency

Definition 1.8

A **tautology** is a statement form where its truth values in all rows in the truth table are always true.

A **contradiction** is a statement form where its truth values in all rows in the truth table are always false.

A **contingency** is a statement form that is neither tautology nor contradiction.

Note

Normally, t is used to denote a tautology and c is used to denote a contradiction.

Example 1.12

Let p , q and r be statement variables. Show that the statement form

- (a) $\sim p \vee p$ is a tautology
- (b) $\sim p \wedge p$ is a contradiction
- (c) $(p \wedge q) \vee \sim r$ is a contingency.

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(a) & (b)

(c)

Conditional Statements

Definition 1.9

If p and q are statements, the statement “if p then q ” or “ p implies q ”, denoted $p \Rightarrow q$, is called a **conditional statement**, or **implication**.

The statement p is called the **hypothesis** and the statement q is called the **conclusion** (or **consequent**).

It is false, when p is true and q is false; otherwise it is true.

Truth Table $p \Rightarrow q$

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Note

(1) A variety of terminology is used to express $p \Rightarrow q$ as given below:

If p then q	q if p
p implies q	q when p
p only if q	q follows from p
p is sufficient for q	q is necessary for p

(2) If the hypothesis p is false, we consider $p \Rightarrow q$ is true regardless of its conclusion.

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$p \Rightarrow q$ is true because p is false is often called **true by default**.

A true conditional statement does not mean a conditional statement with true conclusion.

Example 1.13

Determine the hypothesis and conclusion for each of the following conditional statements. Then determine the truth value.

(1) The moon is square only if the sun rises in the East.

(2) 1 and 3 are prime if 1 multiply 3 is prime.

(3) $(\sin \pi)(\cos \pi) = 0$ when $\sin \pi = 0$ or $\cos \pi = 0$.

(4) If $1 + 1 = 3$, then cats can fly.

Negation, Contrapositive, Converse and Inverse

Definition 1.10

Let p and q be statement variables:

- | | |
|---|-------------------------------|
| (1) The negation of $p \Rightarrow q$ is | $p \wedge \sim q$. |
| (2) The contrapositive of $p \Rightarrow q$ is | $\sim q \Rightarrow \sim p$. |
| (3) The converse of $p \Rightarrow q$ is | $q \Rightarrow p$. |
| (4) The inverse of $p \Rightarrow q$ is | $\sim p \Rightarrow \sim q$. |

Note

Note that the truth tables of $p \Rightarrow q$ and its contrapositive are the same, and the truth tables of the converse and the inverse are the same.

Example 1.14

Write the negation, contrapositive, converse and inverse of the following conditional statements:

(a) If 3 is positive then 3 is nonnegative.

Negation:

Contrapositive:

Converse:

Inverse:

- (b) If you study hard, then you will not fail UTSM3363.

Negation:

Contrapositive:

Converse:

Inverse:

Only if

To say “ p only if q ” means that if q does not take place, then p cannot take place.

Definition 1.11

Let p and q be statement variables, p **only if** q means “if not q then not p ”, or equivalently “if p then q ”.

Biconditional

Definition 1.12

Let p and q be statement variables. The statement form

$$(p \Rightarrow q) \wedge (q \Rightarrow p)$$

Is called the **biconditional of p and q** .

This is read “ **p if and only if q** ” and is denoted $p \Leftrightarrow q$.

It is true if both p and q have the same truth values and it is false if p and q have the opposite truth values.

Truth Table for $p \Leftrightarrow q$

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

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Note

p is a necessary and sufficient condition for q means $p \Leftrightarrow q$.

Example 1.15

Determine the truth values of the following statements:

(a) 2 is prime if and only if it is multiple of 2.

(b) $\sqrt{2}$ is negative if and only if 4 is negative.

(c) $0 < 1 \Leftrightarrow 2 < 1$

Note

- (1) The order of operations can be overridden through the use of parentheses – ().
- (2) The symbol $\wedge$ and $\vee$ are considered coequal in order of operation.
- (3) The order of operations is that $\sim$ is performed first, then $\wedge$ and $\vee$, and finally $\Rightarrow$ and $\Leftrightarrow$.

Exclusive Or

Definition 1.13

The connective corresponding to the **exclusive or** is denoted by $\oplus$.

$p \oplus q$ is true when exactly one of p and q is true.

Note

$p \oplus q$ and $p \Leftrightarrow q$ have opposite truth values.

Logical Equivalence

Definition 1.14

Two statement forms P and Q are called **logically equivalence** if, and only if $P \Leftrightarrow Q$ is a tautology.

The logical equivalence of statement forms P and Q is written as $P \equiv Q$.

Note

One way to determine whether 2 statement forms P and Q are logically equivalent is to construct truth table $P \Leftrightarrow Q$.

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Example 1.16

Let p and q be 2 statement variables. Determine whether the following statement forms are logically equivalent or not.

(a) $\sim p \vee \sim q$ and $\sim(p \vee q)$

(b) $p \vee (p \wedge q)$ and p

Negations of conjunction and disjunction: De Morgan's Law

De Morgan's Law

$$\sim(p \wedge q) \equiv \sim p \vee \sim q$$

$$\sim(p \vee q) \equiv \sim p \wedge \sim q$$

Note

$\sim(p \wedge q)$ is called the **negation of conjunction of p and q** .

$\sim(p \vee q)$ is called the **negation of disjunction of p and q** .

Example 1.17

Write the negation of the given statements:

(1) John is smart but lazy.

(2) $\sqrt{2} \leq 2 < \pi$

Logical equivalences of conditional statements

Theorem 1.1

Let p and q be statements variables.

- (1) $p \Rightarrow q \equiv \sim p \vee q$
- (2) $p \Rightarrow q \equiv \sim q \Rightarrow \sim p$
- (3) $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$

Example 1.18

Prove the logical equivalences

- (1) $p \Rightarrow q \equiv \sim p \vee q$
- (2) $p \Rightarrow q \equiv \sim q \Rightarrow \sim p$

Laws of Logical Equivalences

Theorem 1.2

Given any statement variables p , q and r , a tautology t and a contradiction c , the following logical equivalences hold.

- (1) Commutative laws: $p \wedge q \equiv q \wedge p$
 $p \vee q \equiv q \vee p$
- (2) Associative laws: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
 $(p \vee q) \vee r \equiv p \vee (q \vee r)$
- (3) Distributive laws: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
 $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
- (4) Identity laws: $p \wedge t \equiv p$
 $p \vee c \equiv p$
- (5) Negation laws: $p \vee \sim p \equiv t$
 $p \wedge \sim p \equiv c$
- (6) Double negative laws: $\sim(\sim p) \equiv p$

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- (7) Idempotent laws: $p \wedge p \equiv p$
 $p \vee p \equiv p$
- (8) Universal bound laws: $p \vee \mathbf{t} \equiv \mathbf{t}$
 $p \wedge \mathbf{c} \equiv \mathbf{c}$
- (9) De Morgan's laws: $\sim(p \wedge q) \equiv \sim p \vee \sim q$
 $\sim(p \vee q) \equiv \sim p \wedge \sim q$
- (10) Absorption laws: $p \vee (p \wedge q) \equiv p$
 $p \wedge (p \vee q) \equiv p$
- (11) Negations of $\mathbf{t}$ and $\mathbf{c}$: $\sim \mathbf{t} \equiv \mathbf{c}$
 $\sim \mathbf{c} \equiv \mathbf{t}$

Example 1.19

Show that

- (a) $(p \vee q) \wedge \sim(\sim p \wedge q) \equiv p$
- (b) $\sim[\sim((p \vee q) \wedge r) \vee \sim q] \equiv q \wedge r$

Example 1.20

Simplify the following statement, stating the law used in each step of the simplification:

$$(p \vee q) \wedge p \wedge (r \vee q) \wedge (p \vee \sim p \vee r) \wedge (r \vee \sim q)$$

Rules of Inferences

Definition 1.15

An **argument** is a sequence of statements written

$$\begin{array}{c} p_1 \\ p_2 \\ \vdots \\ p_n \end{array} \quad \text{or } p_1, p_2, \dots, p_n / \therefore q \quad (\therefore \text{ is read as therefore})$$

$$\therefore q$$

which means $p_1 \wedge p_2 \wedge \dots \wedge p_n \Rightarrow q$.

The statements $p_1, p_2, \dots, p_n$ are called **premises** (or **hypothesis** or **assumptions**) and the statement q is called **conclusion**.

Definition 1.16

An argument is **valid** provided that if p_1 and p_2 and ...and p_n are all true, then q must be also true; otherwise the argument is **invalid** (or **fallacy**).

Testing an argument form for validity

- (1) Identity the premises and conclusion
- (2) Construct a truth table showing the truth values of all the premises and the conclusion.
- (3) If the truth table contains a row in which all premises are true and the conclusion is false, then the argument form is invalid.
Otherwise, in every case where all the premises are true, the conclusion is also true, then the argument form is valid.

Example 1.21

Determine the validity of the following argument forms by using truth tables.

$$\begin{array}{l}
 \text{(i)} \quad p \Rightarrow q \vee \sim r \\
 \quad \quad q \Rightarrow p \wedge r \\
 \hline
 \therefore p \Rightarrow r
 \end{array}$$

p	q	r	$\sim r$	$q \vee \sim r$	$p \wedge r$	$p \Rightarrow q \vee \sim r$	$q \Rightarrow p \wedge r$	$p \Rightarrow r$
T	T	T	F	T	T	T	T	T
T	T	F	T	T	F	T	F	F
T	F	T	F	F	T	F	T	T
T	F	F	T	T	F	T	T	F
F	T	T	F	T	F	T	F	T
F	T	F	T	T	F	T	F	T
F	F	T	F	F	F	T	T	T
F	F	F	T	T	F	T	T	T

$$\begin{array}{l}
 \text{(ii)} \quad p \vee (q \vee r) \\
 \quad \quad \sim r \\
 \hline
 \therefore p \vee q
 \end{array}$$

p	q	r	$p \vee (q \vee r)$	$\sim r$	$p \vee q$
T	T	T	T	F	T
T	T	F	T	T	T
T	F	T	T	F	T
T	F	F	T	T	T
F	T	T	T	F	T
F	T	F	T	T	T
F	F	T	T	F	F
F	F	F	F	T	F

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	<u>Rules of Inference</u>	<u>Name</u>
1	$\begin{array}{l} p \Rightarrow q \\ p \\ \hline \therefore q \end{array}$	Modus Ponens (Law of Detachment)
2	$\begin{array}{l} p \Rightarrow q \\ \sim q \\ \hline \therefore \sim p \end{array}$	Modus Tollens
3	$\begin{array}{cc} p & \text{or} & q \\ \hline \therefore p \vee q & & \therefore p \vee q \end{array}$	Disjunctive Addition
4	$\begin{array}{cc} p \wedge q & \text{or} & p \wedge q \\ \hline \therefore p & & \therefore q \end{array}$	Conjunctive Simplification
5	$\begin{array}{cc} p \vee q & \text{or} & p \vee q \\ \sim p & & \sim q \\ \hline \therefore q & & \therefore p \end{array}$	Disjunctive Syllogism
6	$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	Conjunction
7	$\begin{array}{l} p \Rightarrow q \\ q \Rightarrow r \\ \hline \therefore p \Rightarrow r \end{array}$	Hypothetical Syllogism
8	$\begin{array}{l} p \vee q \\ \sim p \vee r \\ \hline \therefore q \vee r \end{array}$	Resolution

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- (6) r from (4) & (5), we use modus ponens
(7) $p \Rightarrow s$ from (1) & (3), we use hypothetical syllogism
(8) s from (4) & (7), we use modus ponens
(9) $r \wedge s$ from (6) & (8), we use conjunction
So we conclude that the conclusion does follow from the hypotheses

Example 1.25

Show that the hypotheses

“If John takes the computer course, then John stays in the hostel”

“John does not stay in the hostel”

“If John does not take the computer course, then John takes the language course or stay at home”

“If John takes language course then John buys a motorcycle”

“If John buys a car, then John does not buy motorcycle”

“John has a car”

lead to the conclusion “John stays at home”

Let

p = John takes the computer course

q = John stays in the hostel

r = John takes the language course

s = John stays at home

t = John buys a motorcycle

u = John buys a car

Converse Error

Definition 1.17

Modus Ponens states that $((p \Rightarrow q) \wedge p) \Rightarrow q$.

The argument $((p \Rightarrow q) \wedge q) \Rightarrow p$ is invalid and is called the **converse error** because the conclusion of the argument would follow from the premises if the premises $p \Rightarrow q$ were replaced by its converse. Such a replacement is not allowed, however, because a conditional statement is not logically equivalent to its converse.

Inverse Error

Definition 1.18

Modus Tollens states that $((p \Rightarrow q) \wedge \sim q) \Rightarrow \sim p$.

The argument $((p \Rightarrow q) \wedge \sim p) \Rightarrow \sim q$ is invalid and is called the **inverse error** because the conclusion of the argument would follow from the premises if the premise $p \Rightarrow q$ were replaced by its inverse. Such a replacement is not allowed, however, because a conditional statement is not logically equivalent to its inverse.

Example 1.26

Are the following arguments valid?

- (1) If Ali is tall, then he sits in the back row.

Ali sits in the back row.

Therefore Ali is tall

- (2) If interest rates are going up, stock market prices will go down.

Interest rates are not going up.

Therefore stock market prices will not go down.

Example 1.27

Determine the validity of the following arguments:

$$\begin{array}{l} \text{(a)} \quad p \Rightarrow q \\ \quad \quad \sim p \Rightarrow r \\ \quad \quad \hline \quad \quad r \Rightarrow s \\ \therefore \sim q \Rightarrow s \end{array}$$

$$\begin{array}{l} \text{(b)} \quad (\sim p \vee \sim q) \Rightarrow (r \wedge s) \\ \quad \quad r \Rightarrow t \\ \quad \quad \hline \quad \quad \sim t \\ \therefore p \end{array}$$